

100 MCQs with Answers

1. Force is basically a:

- a) Push or pull
- b) Sound or light
- c) Heat or pressure
- d) Weight or mass

Answer: a) Push or pull

2. Motion means:

- a) Object not changing position
- b) Object changing its position with time
- c) Object remaining still
- d) Object becoming heavier

Answer: b) Object changing its position with time

3. An object is said to be at rest if:

- a) It moves continuously
- b) It changes its shape
- c) It does not change its position with respect to surroundings
- d) It rotates

Answer: c) It does not change its position with respect to surroundings

4. A book lying on a table is an example of:

- a) Motion
- b) Rest
- c) Force
- d) Friction

Answer: b) Rest

5. A car parked in a parking area is an example of:

- a) Motion
- b) Rest
- c) Friction
- d) Energy

Answer: b) Rest

6. A man bicycling on a road is an example of:

- a) Rest
- b) Motion
- c) Force
- d) Friction

Answer: b) Motion

7. A leaf falling from a tree shows:

- a) Rest
- b) Motion
- c) Heat
- d) Light

Answer: b) Motion

8. The push or pull on an object is called:

- a) Energy
- b) Power
- c) Force
- d) Speed

Answer: c) Force

9. Force acts when:

- a) Two objects are in contact
- b) There is no object
- c) The object is far away
- d) There is light

Answer: a) Two objects are in contact

10. The direction in which a force is applied is called:

- a) Application point
- b) Direction of force
- c) Motion
- d) Rest

Answer: b) Direction of force

11. The point where the force is applied is called:

- a) Focus point
- b) Centre point
- c) Application of force
- d) Rest point

Answer: c) Application of force

12. Force is measured using a:

- a) Thermometer
- b) Spring balance
- c) Barometer
- d) Stopwatch

Answer: b) Spring balance

13.The SI unit of force is:

- a) Joule
- b) Watt
- c) Newton
- d) Pascal

Answer: c) Newton

14.Force has:

- a) Only magnitude
- b) Only direction
- c) Both magnitude and direction
- d) Neither magnitude nor direction

Answer: c) Both magnitude and direction

15.Contact forces act:

- a) Without touching
- b) Through a medium or direct contact
- c) From a distance
- d) In space only

Answer: b) Through a medium or direct contact

16.Non-contact forces act:

- a) Through direct contact
- b) Through distance without contact
- c) Only on humans
- d) Only on solids

Answer: b) Through distance without contact

17.Muscular force is an example of:

- a) Contact force
- b) Non-contact force
- c) Electrostatic force
- d) Gravitational force

Answer: a) Contact force

18.Frictional force acts between:

- a) Air and light
- b) Surfaces in contact
- c) Water and fire
- d) Heat and cold

Answer: b) Surfaces in contact

19.Friction always:

- a) Helps motion
- b) Opposes motion
- c) Speeds up motion

d) Doubles motion

Answer: b) Opposes motion

20. Mechanical force is a:

- a) Contact force
- b) Non-contact force
- c) Gravitational force
- d) Electrostatic force

Answer: a) Contact force

21. The gravitational force acts between:

- a) Two magnets
- b) Earth and objects
- c) Two charged bodies
- d) Two wires

Answer: b) Earth and objects

22. The magnetic force acts between:

- a) Two stones
- b) Two magnets
- c) Two humans
- d) Two leaves

Answer: b) Two magnets

23. Electrostatic force acts between:

- a) Moving vehicles
- b) Charged bodies
- c) Magnets
- d) Water and air

Answer: b) Charged bodies

24. Which of these is a non-contact force?

- a) Friction
- b) Muscular force
- c) Gravitational force
- d) Mechanical force

Answer: c) Gravitational force

25. The force exerted by the earth on all objects is called:

- a) Friction
- b) Gravity
- c) Magnetism

d) Pressure

Answer: b) Gravity

26. The force responsible for falling objects is:

- a) Magnetic force
- b) Gravitational force
- c) Frictional force
- d) Elastic force

Answer: b) Gravitational force

27. The weight of a body is the force:

- a) Pushed by air
- b) Pulled by the earth towards its center
- c) Pulled by the sun
- d) Pulled by the moon

Answer: b) Pulled by the earth towards its center

28. Motion is when:

- a) Object stays still
- b) Object changes position over time
- c) Object is heavy
- d) Object is cold

Answer: b) Object changes position over time

29. A soccer ball rolling on a field shows:

- a) Rest
- b) Motion
- c) Friction
- d) Rotation

Answer: b) Motion

30. A stationary bicycle is an example of:

- a) Motion
- b) Rest
- c) Friction
- d) Speed

Answer: b) Rest

31. Linear motion occurs when:

- a) Object moves in a straight line
- b) Object moves in a circle
- c) Object rotates
- d) Object vibrates

Answer: a) Object moves in a straight line

32. A person walking on a straight path shows:

- a) Curvilinear motion

- b) Linear motion
- c) Circular motion
- d) Rotational motion

Answer: b) Linear motion

33. Curvilinear motion occurs when:

- a) Object moves in a straight path
- b) Object moves in a curved path
- c) Object is stationary
- d) Object spins in place

Answer: b) Object moves in a curved path

34. A roller coaster moving on a track shows:

- a) Linear motion
- b) Curvilinear motion
- c) Rotational motion
- d) Oscillatory motion

Answer: b) Curvilinear motion

35. Circular motion occurs when:

- a) Object moves around a fixed point
- b) Object moves in a zigzag line
- c) Object moves randomly
- d) Object stays still

Answer: a) Object moves around a fixed point

36. Swinging a yo-yo in a circle shows:

- a) Linear motion
- b) Circular motion
- c) Rotational motion
- d) Oscillatory motion

Answer: b) Circular motion

37. A ceiling fan shows:

- a) Rotational motion
- b) Linear motion
- c) Oscillatory motion
- d) Curvilinear motion

Answer: a) Rotational motion

38. The blades of a fan move around an:

- a) Axis
- b) Straight line
- c) Fixed square
- d) Triangle

Answer: a) Axis

39. A potter's wheel is an example of:

- a) Oscillatory motion
- b) Rotational motion
- c) Linear motion
- d) Curvilinear motion

Answer: b) Rotational motion

40. Oscillatory motion means:

- a) Back and forth movement
- b) Straight movement
- c) Circular movement
- d) Random movement

Answer: a) Back and forth movement

41. The motion of a pendulum is:

- a) Circular
- b) Oscillatory
- c) Linear
- d) Rotational

Answer: b) Oscillatory

42. The motion of a sewing machine needle is:

- a) Rotational
- b) Oscillatory
- c) Circular
- d) Curvilinear

Answer: b) Oscillatory

43. Zigzag motion means:

- a) Straight motion
- b) Motion in different directions
- c) Circular motion
- d) No motion

Answer: b) Motion in different directions

44. People walking in a crowded street show:

- a) Circular motion
- b) Zigzag motion
- c) Linear motion
- d) Oscillatory motion

Answer: b) Zigzag motion

45. Periodic motion means:

- a) Repeating motion in regular intervals
- b) Irregular motion
- c) Random movement

d) One-time movement

Answer: a) Repeating motion in regular intervals

46. The motion of a swing is:

a) Periodic motion

b) Rotational motion

c) Linear motion

d) Curvilinear motion

Answer: a) Periodic motion

47. Non-periodic motion means:

a) Regular repeating motion

b) Motion not repeating at regular intervals

c) Motion around a circle

d) Motion in a straight line

Answer: b) Motion not repeating at regular intervals

48. Driving a car in traffic is an example of:

a) Periodic motion

b) Non-periodic motion

c) Oscillatory motion

d) Circular motion

Answer: b) Non-periodic motion

49. Uniform motion means:

a) Covering equal distances in equal intervals of time

b) Changing distance randomly

c) Motion that stops suddenly

d) Irregular speed

Answer: a) Covering equal distances in equal intervals of time

50. Non-uniform motion means:

a) Motion with constant speed

b) Motion with changing speed

c) Motion in a straight line

d) Motion with no change

Answer: b) Motion with changing speed

51. What is defined as a push or pull acting on an object?

a) Energy

b) Power

c) Force

d) Pressure

Answer: c) Force

52. Which of the following is NOT a contact force?

a) Frictional force

- b) Muscular force
- c) Magnetic force
- d) Mechanical force

Answer: c) Magnetic force

53. The SI unit of force is —

- a) Joule
- b) Newton
- c) Watt
- d) Pascal

Answer: b) Newton

54. A book lying on a table is an example of —

- a) Motion
- b) Rest
- c) Friction
- d) Force

Answer: b) Rest

55. A car moving on a road is an example of —

- a) Rest
- b) Motion
- c) Friction
- d) Pressure

Answer: b) Motion

56. The instrument used to measure force is —

- a) Thermometer
- b) Spring balance
- c) Barometer
- d) Stopwatch

Answer: b) Spring balance

57. Which force acts between two surfaces in contact and opposes motion?

- a) Magnetic force
- b) Electrostatic force
- c) Gravitational force
- d) Frictional force

Answer: d) Frictional force

58. Which of the following is an example of a non-contact force?

- a) Frictional force
- b) Gravitational force
- c) Muscular force
- d) Mechanical force

Answer: b) Gravitational force

59. Which type of force is used when you push a door to open it?

- a) Muscular force
- b) Gravitational force
- c) Magnetic force
- d) Electrostatic force

Answer: a) Muscular force

60. The direction in which force is applied is called —

- a) Point of application
- b) Direction of the force
- c) Force line
- d) None of these

Answer: b) Direction of the force

61. Which force is responsible for objects falling to the ground?

- a) Electrostatic force
- b) Magnetic force
- c) Gravitational force
- d) Frictional force

Answer: c) Gravitational force

62. The weight of a body is the —

- a) Force exerted by the body on earth
- b) Force of earth pulling the body towards its center
- c) Energy possessed by the body
- d) None of these

Answer: b) Force of earth pulling the body towards its center

63. Which of the following is an example of mechanical force?

- a) Horse pulling a cart
- b) Earth attracting objects
- c) Magnet attracting iron
- d) Plastic rubbed with wool attracting paper bits

Answer: a) Horse pulling a cart

64. When a body does not change its position with respect to its surroundings, it is said to be —

- a) In motion
- b) At rest
- c) Rotating
- d) Oscillating

Answer: b) At rest

65. The change in position of a body with time is known as —

- a) Rest

- b) Motion
- c) Friction
- d) Gravity

Answer: b) Motion

66. Which motion involves moving along a straight path?

- a) Circular motion
- b) Linear motion
- c) Oscillatory motion
- d) Rotational motion

Answer: b) Linear motion

67. A rolling soccer ball is an example of —

- a) Rest
- b) Motion
- c) Friction
- d) Rotation

Answer: b) Motion

68. The motion of a pendulum is an example of —

- a) Rotational motion
- b) Zigzag motion
- c) Oscillatory motion
- d) Linear motion

Answer: c) Oscillatory motion

69. Which motion is shown by a fan's blades?

- a) Rotational motion
- b) Linear motion
- c) Circular motion
- d) Periodic motion

Answer: a) Rotational motion

70. Which type of motion is shown by a person walking in a crowded street?

- a) Linear motion
- b) Zigzag motion
- c) Periodic motion
- d) Rotational motion

Answer: b) Zigzag motion

71. When an object moves around a fixed point, it shows —

- a) Linear motion

- b) Circular motion
- c) Oscillatory motion
- d) Non-periodic motion

Answer: b) Circular motion

72. A car moving on a curved road shows —

- a) Curvilinear motion
- b) Linear motion
- c) Oscillatory motion
- d) Rotational motion

Answer: a) Curvilinear motion

73. The up and down motion of a sewing machine needle is —

- a) Oscillatory motion
- b) Rotational motion
- c) Linear motion
- d) Periodic motion

Answer: a) Oscillatory motion

74. The repeated motion that occurs at equal intervals of time is called —

- a) Zigzag motion
- b) Periodic motion
- c) Non-periodic motion
- d) Irregular motion

Answer: b) Periodic motion

75. The movement of a car in traffic is an example of —

- a) Uniform motion
- b) Periodic motion
- c) Non-periodic motion
- d) Rotational motion

Answer: c) Non-periodic motion

76. Motion that repeats itself at regular intervals of time is —

- a) Non-uniform
- b) Periodic
- c) Random
- d) Uniform

Answer: b) Periodic

77. The motion that does not repeat at regular intervals is —

- a) Periodic motion
- b) Non-periodic motion
- c) Rotational motion
- d) Circular motion

Answer: b) Non-periodic motion

78. The equation for speed is —

- a) $\text{Speed} = \text{Distance} \times \text{Time}$
- b) $\text{Speed} = \text{Distance} / \text{Time}$
- c) $\text{Speed} = \text{Time} / \text{Distance}$
- d) $\text{Speed} = \text{Force} \times \text{Time}$

Answer: b) Speed = Distance / Time

79. If an object covers equal distances in equal intervals of time, it is said to be in —

- a) Uniform motion
- b) Non-uniform motion
- c) Rotational motion
- d) Irregular motion

Answer: a) Uniform motion

80. The SI unit of speed is —

- a) m/s
- b) km
- c) s/m
- d) m^2/s

Answer: a) m/s

81. Which type of motion is shown by a swing in a playground?

- a) Linear motion
- b) Periodic motion
- c) Rotational motion
- d) Circular motion

Answer: b) Periodic motion

82. When an object's speed changes with time, its motion is —

- a) Uniform
- b) Non-uniform
- c) Periodic
- d) Linear

Answer: b) Non-uniform

83. A ceiling fan shows —

- a) Rotational motion
- b) Linear motion
- c) Oscillatory motion
- d) Zigzag motion

Answer: a) Rotational motion

84. A car parked in a parking area is an example of —

- a) Rest
- b) Motion

c) Friction

d) Rotation

Answer: a) Rest

85. The motion of a clock pendulum is —

a) Linear motion

b) Oscillatory motion

c) Rotational motion

d) Non-periodic motion

Answer: b) Oscillatory motion

86. The magnetic force acts —

a) Only on contact

b) Through a distance

c) Only in water

d) Only on heavy objects

Answer: b) Through a distance

87. The force exerted by the Earth on objects is known as —

a) Magnetic force

b) Frictional force

c) Gravitational force

d) Electrostatic force

Answer: c) Gravitational force

88. What is the full form of SI?

a) Standard International

b) System of International Units

c) Scientific Index

d) Standard of India

Answer: b) System of International Units

89. When you stretch a rubber band, which type of force is applied?

a) Frictional

b) Muscular

c) Electrostatic

d) Magnetic

Answer: b) Muscular

90. What type of motion is shown by a potter's wheel?

a) Circular

b) Rotational

c) Oscillatory

d) Periodic

Answer: b) Rotational

91.The movement of the Earth around the Sun is an example of —

- a) Linear motion
- b) Circular motion
- c) Rotational motion
- d) Zigzag motion

Answer: b) Circular motion

92.A body that changes its position with respect to time is said to be —

- a) At rest
- b) In motion
- c) Stationary
- d) Rotating

Answer: b) In motion

93.A football rolling on the ground stops due to —

- a) Gravity
- b) Friction
- c) Magnetism
- d) Air pressure

Answer: b) Friction

94.The to-and-fro motion of a swing is —

- a) Linear
- b) Circular
- c) Oscillatory
- d) Zigzag

Answer: c) Oscillatory

95.The force which keeps the moon in its orbit around the Earth is —

- a) Magnetic force
- b) Electrostatic force
- c) Gravitational force
- d) Muscular force

Answer: c) Gravitational force

96.If a car travels 100 km in 2 hours, its speed is —

- a) 25 km/h
- b) 50 km/h
- c) 75 km/h
- d) 100 km/h

Answer: b) 50 km/h

97.A pendulum completes one swing in 2 seconds. This time is called —

- a) Interval
- b) Period
- c) Frequency

d) Cycle

Answer: b) Period

98. A magnet attracting iron nails is an example of —

- a) Frictional force
- b) Non-contact force
- c) Contact force
- d) Muscular force

Answer: b) Non-contact force

99. The motion of the Earth spinning on its axis is —

- a) Rotational motion
- b) Circular motion
- c) Oscillatory motion
- d) Linear motion

Answer: a) Rotational motion

100. When force is applied on an object, it can —

- a) Only stop motion
- b) Only change shape
- c) Change motion or shape
- d) Only increase weight

Answer: c) Change motion or shape